Monologue or Dialogue?

Measuring Conversational Structure in the Reasoning Traces of Large Language Models

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Abstract

A reasoning model such as DeepSeek-R1 produces a long "thinking" trace before it gives an answer. That trace can be read as one mind working step by step, or as several voices exchanging proposals and objections. This paper asks which reading the text itself supports once it is measured. No reasoning trace comes labelled with its true voice structure, so we build a small grid of corpora whose structure is known: human and AI, monologue and dialogue. We check that each of three text-analysis methods can separate monologue from dialogue on these known corpora, and then apply them to the traces. Two of the three place the traces on the dialogue side of the scale, and the third, which reads the coarsest signal, places them just onto the monologue side by a small margin. The three methods are not equally well validated, so we state how much weight each one can bear and flag the one that depends on an automatic labeller. Taken together, the traces show several surface marks of dialogue, most clearly in how the writing style shifts from sentence to sentence and in how often a question is followed by an answer.

1 The question

Reasoning-oriented training has become standard in frontier models. Given a problem, a model such as DeepSeek-R1 [1] first produces a long stretch of intermediate text, often called a reasoning trace [2], and only then gives its final answer. These traces have a notable feature: each is written by a single model in one pass, yet a trace often contains questions and their answers, objections and concessions, and sudden reversals such as "Wait, that is not right." In writing by people, features like these can mark a real exchange between participants, though they can also mark a single inner voice. This raises a question that is easy to pose and hard to settle: does a reasoning trace have the textual shape of a monologue or of a dialogue?

The question also bears on current theory. Kim et al. [3] argue that reasoning-trained models improve in part because they simulate a "society of thought," a set of internal perspectives that propose, criticise, and reconcile. Older ideas in cognitive science point the same way, from reasoning as a social and argumentative activity [4] to the mind as a set of interacting agents [5]. Our premise is narrower and testable: if a trace is produced by several voices taking turns, that should leave marks in the writing. Writers differ in small and largely involuntary habits, such as how they use function words, how their sentences flow, and which conversational moves they make, and authorship analysis and conversation analysis use these habits to tell writers and turns apart. We therefore treat the question in operational terms: a monologue should read as the work of one author, and a dialogue as several voices exchanging turns, and we ask which of these the traces resemble.

2 The design: a calibration grid

No collection of reasoning traces comes labelled with its true voice structure, which is what makes the question hard to answer directly. We address this with the grid in Figure 1. We assemble four corpora whose structure is known and vary two properties across them: who produced the text (a human or an AI) and whether it is a monologue or a dialogue. The four cells are human monologue (think-aloud protocols, in which a person speaks their thoughts aloud while solving a task [6]), human dialogue (group deliberation from DeliData [7]), AI monologue (a single Qwen-2.5-7B model solving a puzzle on its own), and AI dialogue (four separate copies of Qwen-2.5-7B, each given a role, taking turns in a debate). The fifth cell, the DeepSeek-R1 traces, is the one whose structure we want to determine.

The grid functions as a validity check, or gate. A method may report on the traces only after it has shown that it can separate monologue from dialogue on all four known corpora. The rationale is straightforward: a method that cannot tell the voices apart in a genuine four-way debate is in no position to rule them out in a trace, and a method that reports several voices in one person's think-aloud protocol is in no position to affirm them there.

One design choice keeps the comparison clean: we hold the task fixed where possible. The AI monologue, AI dialogue, and DeepSeek-R1 cells all solve the same puzzle, Game of 24 [8], a small arithmetic puzzle in which four given numbers must be combined with the four basic operations to reach 24, and the human monologue cell uses the same puzzle. A method that reads a monologue as a single voice therefore cannot be responding merely to the subject matter. The human dialogue cell is the one exception, since it uses the Wason card task, another compact logic puzzle, because no group-deliberation corpus exists for Game of 24; its role is only to confirm that a method can detect human dialogue at all, so it never anchors the placement. We also removed speaker names, role titles, and letter labels from both dialogue corpora before analysis, so that a method cannot detect dialogue by identifying who is speaking instead of reading the structure of the exchange.

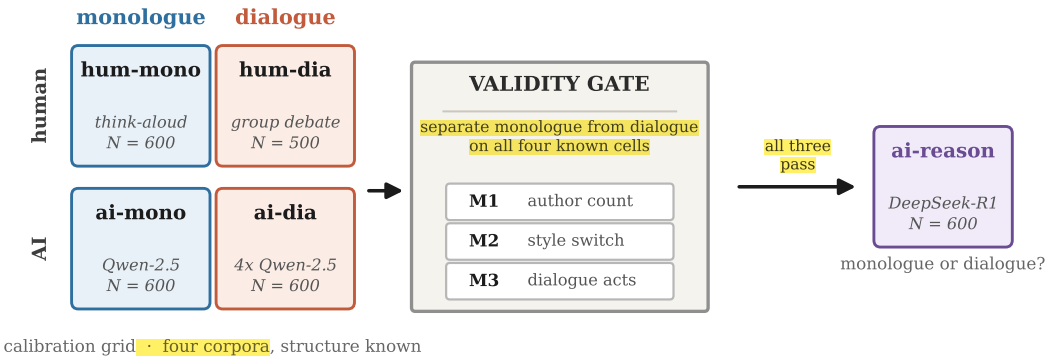


Figure 1: Study design. Four corpora of known voice structure (left) feed a validity check that each method must pass: separating monologue from dialogue on all four known cells. Methods that pass are applied to the DeepSeek-R1 traces (`ai-reason`) and place them on a monologue–dialogue scale anchored by the two same-task AI cells; the human cells serve only to confirm that a method works. The check shown here is the monologue/dialogue separation, which all three methods pass. Method 3 faces a further check on the reliability of its labels, described in Section 4.

3 Three methods, three signals

The three methods read deliberately different signals, so that agreement among them carries more weight than any single method alone.

Method 1: counting authors across the document [9]. This method treats a document as a whole. It divides the text into paragraph-sized windows, describes each window by how often the 50 most common words occur in it, and groups similar windows together. The number of groups is read as the number of authors, which we reduce to a binary decision: one author, or more than one. Its check is balanced accuracy across the four known cells, which must exceed the level expected by chance. Balanced accuracy here is the average of two quantities, how often the method is correct on the single-author cells and how often it is correct on the multi-author cells, which prevents an uneven mix of the two classes from inflating the score.

Method 2: detecting style changes between sentences [10]. This method operates sentence by sentence. For each pair of neighbouring sentences it estimates the probability that the two were written by different authors, using an encoder trained to capture writing style while ignoring topic [11]. Longer documents contain more sentence pairs, which affects the raw score, so we adjust for length before comparing cells. Each document receives a single number, which we call its *lean*, indicating whether its style-change pattern lies closer to the dialogue cells or to the monologue cells. Its check is that the dialogue cells must lean more toward dialogue than the monologue cells, by a clear margin (Cliff's δ of at least 0.33; Cliff's δ is a rank-based measure of how cleanly two groups separate, with a size of 0 when they overlap completely and 1 when they do not overlap at all).

Method 3: measuring question-and-answer structure [12, 13]. This method reads the text as a sequence of conversational moves. A language model tags each utterance with its function using CAMS, a standard scheme whose labels include a question, an answer, an agreement, or a suggestion. From these tags we measure how often an *adjacency pair* is completed. An adjacency pair is a two-part exchange in which the first part makes the second one due, such as a question followed by an answer, or a proposal followed by its acceptance. A single writer has little reason to produce completed pairs, so their presence is evidence of genuine dialogue. Method 3 must clear two checks, and is the only method to which this applies. The first is the same check the others face: its completion rate must separate the dialogue cells from the monologue cells. The second applies to Method 3 alone, because its labels are produced by a language model acting as a tagger. For that check we compare the tagger's labels with a human annotator's and require sufficient agreement.

4 Results

Method 1 passes its check, but by a narrow margin. Its balanced accuracy is 0.6088 (95% confidence interval 0.596 to 0.621), above the chance level of 0.50. The composition of this score is important. On single-author text the method is nearly perfect, correctly labelling 0.975 of the human monologue documents and 0.998 of the AI monologue documents. On multi-author text it is far weaker, recovering 0.423 of the AI debates and none of the human debates. In sum, the method confirms a single author reliably but detects multiple authors poorly. A monologue-side placement therefore indicates the absence of a detected authorship signal, and should not be taken as positive evidence of a single voice. This caveat applies to its verdict on the traces.

Method 2 passes cleanly. The dialogue cells lean more toward dialogue than the monologue cells (average lean 0.0125 versus −0.0105), and the separation is strong, with Cliff's $\delta = 0.6211$, well above the 0.33 floor. The traces themselves lean further toward dialogue than either AI anchor.

Method 3 requires closer examination, since it must clear two checks. On the first check, which bears directly on our question, it performs well. Its completion rate separates dialogue from monologue in the expected direction (Cliff's $\delta = 0.4717$, above the 0.33 floor). Within each producer, the dialogue cell completes more pairs than the monologue cell: 0.235 against 0.096 for the human cells, and 0.025 against 0.003 for the AI cells. The tagger therefore performs the essential task, distinguishing genuine dialogue from genuine monologue.

On the second check, the tagger falls short of the threshold set in advance. We compared its labels with a human annotator's and measured agreement using Cohen's κ, an agreement score that corrects for the rate at which two labellers would match by chance. The tagger reached $\kappa = 0.54$, below the required 0.70.

We report Method 3's result despite this outcome, and explain the decision here, since it departs from our original plan. Two considerations motivate it. The first is the separation just described: the check that bears directly on our question, distinguishing monologue from dialogue, is the one the tagger passed, and passed clearly. The second is that the 0.70 threshold is too strict for this labelling scheme. CAMS uses a set of 28 fine-grained labels, many of them close relatives, such as `inform` and `suggest`, which are easily confused even by careful human coders. Unweighted κ penalises a near miss between two similar tags as heavily as a disagreement between two unrelated ones. The CAMS authors make the same point and recommend an agreement measure that gives partial credit for near misses. Under such a measure, agreement rises to a tree-weighted value of 0.62, and the raw share of labels on which tagger and human agree is 0.74. These values remain below our original threshold, so we do not claim the labels are error-free. They indicate that the labels are reasonable, and that the pre-registered threshold was too severe for an inherently ambiguous scheme.

To make this dependence explicit, we state one assumption and carry it through the paper: wherever a result relies on Method 3, it also relies on the assumption that the tagger is reliable. We state this at each point, shade Method 3 in gray in every figure, and mark it with a dagger in every table. Readers who accept the assumption may weigh Method 3 alongside the other two; readers who do not may set it aside, since the main result rests on Methods 1 and 2 alone. Under the assumption, Method 3's completion rate places the traces on the dialogue side: the trace rate of 0.035 lies closer to the AI dialogue anchor (0.025) than to the AI monologue anchor (0.003).

Across the three methods (Table 1 and Figure 2), two of the signals place the traces on the dialogue side and one places them just onto the monologue side. Method 2, reading style changes between sentences, places them firmly past the dialogue anchor. Method 3, reading completed question-and-answer pairs, places them past the dialogue anchor as well, under the assumption above. Method 1, which reads only the coarse, whole-document sense of authorship, is the method that leans monologue, and it does so only marginally, at the edge of its no-decision band. The two methods that read fine-grained, turn-by-turn structure agree; the dissenting method is the one that reads the least detail.

Table 1: Placement of the reasoning traces by each method. The *mono/dia* column is the validity check every method must pass, separating monologue from dialogue on the known corpora. The *labels* column is a further check that applies only to Method 3, on the reliability of its automatic tags; Method 3 did not pass it. Method 3's verdict is therefore conditional (†) and shown in gray.

Method	mono/dia	labels	Verdict
1: author count	pass	n/a	monologue (marginal)
2: style switch	pass	n/a	dialogue
3: adjacency pairs	pass	fail	dialogue †

5 What it means

The claim here concerns the surface of the text, that is, the patterns a reader or a tool can observe in the writing. It says nothing directly about what occurs inside the model. On that surface, the picture leans toward dialogue. The traces change writing style from sentence to sentence in the manner of dialogue, and, under the valid-tagger assumption, they complete question-and-answer pairs at a rate above the dialogue anchor. The one method that points the other way reads the coarsest signal and makes only a marginal call; it is a conservative detector, so its monologue reading indicates the absence of an authorship signal, and should not be read as positive evidence of a single voice.

The scope of this claim is limited. A dialogue-leaning reading does not show that a model contains several minds, just as a novel full of dialogue does not have several authors. It shows that the trace is organised like a conversation, which is a measurable fact in itself, and a natural surface counterpart to the internal multi-agent behaviour that Kim et al. [3] describe from the inside, through interpretability. The outside view and the inside view point the same way, which is reassuring, though the present evidence stands on its own as a statement about surface structure.

The study supports two readings, depending on how far the tagger is trusted. Under the valid-tagger assumption, two independent structural signals agree that the traces are dialogue-like, and the case is fairly strong. Without that assumption, the evidence rests on Methods 1 and 2, which disagree: one marginal monologue call from a coarse detector, and one clear dialogue call from a finer one. Even in the more cautious reading, nothing points cleanly to monologue, and the sharpest single signal points to dialogue. We present both readings so that they can be weighed separately.

Two caveats temper the dialogue placements, and both concern how a distance was measured rather than its direction. Method 2's distance to the monologue side is partly an extrapolation, because the traces are much longer than the monologue anchor, so it is more accurate to say that they track the dialogue anchor closely over the range where the two overlap. Method 3's verdict rests on its completion signal, since its coarser tag-frequency profiles cannot separate the two AI anchors. Neither caveat changes a direction; each limits only how far past the pole the placement should be read.

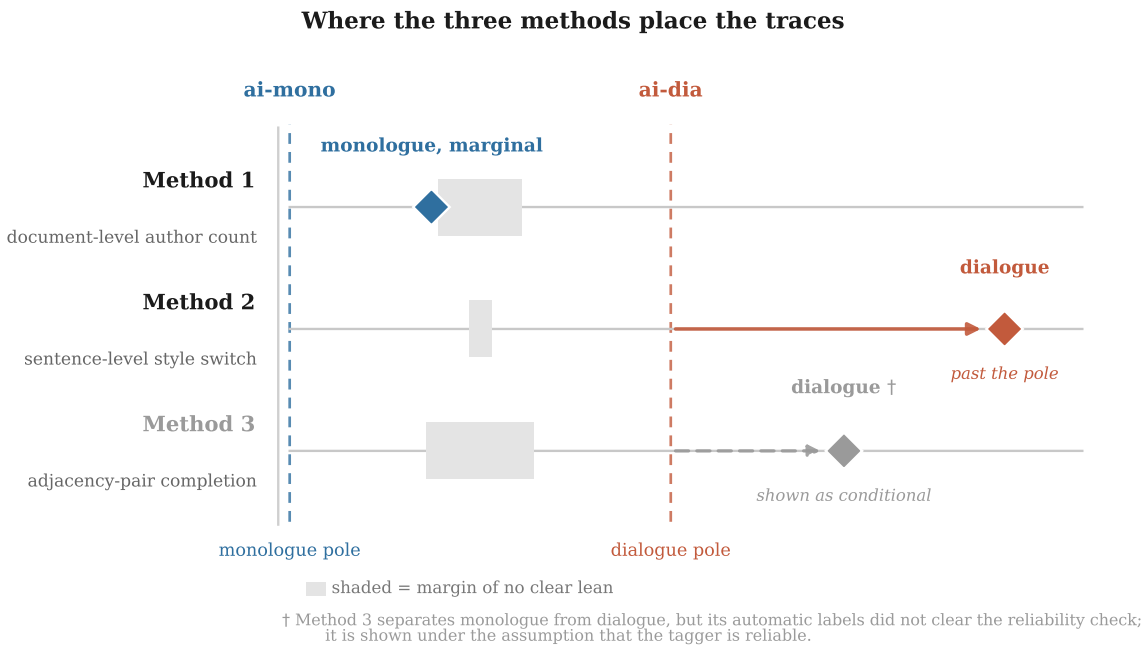


Figure 2: Placement of the traces by each method on the monologue–dialogue scale (poles at 0 and 1). The shaded band is each method's margin, within which no lean is reported. Method 1 sits at the edge of a wide band, just onto the monologue side, a marginal call. Method 2 sits well past the dialogue pole. Method 3 also sits past the dialogue pole, and is shown in gray with a dagger to indicate that it depends on the valid-tagger assumption of Section 4.

6 Limitations

Several limitations should be noted. The target is a single reasoning model, DeepSeek-R1, on a single family of puzzles, so the verdict concerns this setting and does not yet generalise to reasoning traces at large. The two anchors were generated with a different base model (Qwen-2.5-7B) than the target, so part of the measured distance may reflect the change of model as well as voice structure, and the placement is best read as "closer to which pole" than as an exact distance. Method 2's monologue comparison rests on extrapolation, as noted above. Method 3's labels come from an automatic tagger, which is why we state the valid-tagger assumption explicitly and mark its results throughout. The validity check is the basis for our confidence: a method that clears it has shown that it can distinguish the two structures on data whose labels are known.

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